

# BIKE SALES IN EUROPE (2011-2016) INSIGHT REPORT

**Name: Lavender Achieng' Onyango-671809**

## 1. Dashboard Overview

The dashboard presents an analysis of bike sales in Europe between 2011 and 2016. It summarizes key financial metrics, product performance, customer demographics, geographical sales distribution, and sales trends. The dashboard enables decision-makers to monitor business performance and identify opportunities for growth.

## 2. Dashboard Output

The dashboard reports the following key performance indicators (KPIs):

| KPI           | Value          |
|---------------|----------------|
| Total Revenue | €84.83 Million |
| Total Profit  | €32.05 Million |
| Total Cost    | €30.01 Million |

The dashboard also includes:

- Revenue by Product Category
- Revenue Trend (2011–2016)
- Revenue by Age Group
- Profit by Country
- Unit Cost by Product
- Profit Distribution by Country
- Revenue by Country and Product Category
- Sales Details Table
- Geographic Distribution of Profit

### 3. Analysis and Business Insights

#### 1. Strong Business Performance

The company generated **€84.83 million** in revenue while earning **€32.05 million** in profit. This indicates healthy profitability, with profit representing a substantial share of total revenue.

##### Explanation

The difference between revenue and cost demonstrates that the business has been able to maintain positive margins across the reporting period. This suggests efficient pricing and cost management.

#### 2. Bikes Generate the Highest Revenue

The **Revenue by Product Category** chart shows that **Bikes** contribute significantly more revenue than Accessories and Clothing.

##### Explanation

Customers spend considerably more on bicycles than on other product categories. Accessories and Clothing make relatively small contributions to total sales.

##### Business implication

The company should continue investing in bicycle inventory, marketing, and product innovation while exploring opportunities to increase accessory and clothing sales through promotions or bundled offers.

#### 3. Revenue Increased Over Time

The **Revenue Trend** indicates that revenue generally increased between **2011 and 2015**, reaching its highest level around **2015**, before showing a slight decline in **2016**.

### **Explanation**

The upward trend suggests increasing customer demand and business growth over most of the period. The decline in the final year may indicate changing market conditions, increased competition, or incomplete yearly data.

## **4. Adults Generate the Highest Revenue**

The **Revenue by Age Group** chart shows that **Adults** contribute the largest share of revenue, followed by Young Adults, while Seniors contribute the least.

### **Explanation**

Adults appear to be the company's primary customer segment and therefore represent the most valuable market.

### **Business implication**

Marketing campaigns should continue targeting adults while developing strategies to attract younger and older customer groups.

## **5. Profit Varies Across Countries**

The **Profit by Country** chart shows noticeable differences in profitability among countries.

The dashboard suggests that countries such as **the United States** and **Australia** contribute higher profits than several other regions.

### **Explanation**

Some markets are significantly more profitable than others due to higher sales volumes, pricing strategies, or stronger customer demand.

### **Business implication**

The company should identify successful practices in high-performing countries and replicate them in lower-performing markets where feasible.

## **6. Geographic Distribution Highlights Major Markets**

The map visual indicates that profit is concentrated in a few major countries.

### **Explanation**

Sales are not evenly distributed geographically. Certain countries contribute substantially more to company performance.

### **Business implication**

Future expansion efforts should focus on regions with growing demand while improving performance in underperforming markets.

## **7. Product Costs Differ Significantly**

The **Total Unit Cost by Product** chart shows that some bike models have much higher unit costs than others.

### **Explanation**

Premium bicycles require higher production costs but may also generate greater profit if appropriately priced.

## **4. Challenges Encountered**

Several challenges were experienced during the development of the dashboard:

- Cleaning inconsistent text values and ensuring data quality in Power Query.
- Selecting appropriate data types for dates and numerical variables.
- Designing a dashboard layout that remained readable despite the number of required visuals.
- Choosing appropriate colors that maintain readability while presenting information professionally.
- Configuring the map visual correctly by ensuring the Country field was categorized as a geographic location.
- Balancing multiple visuals without overcrowding the report page.

## **5. Recommendations**

Based on the dashboard analysis, the following recommendations are proposed:

1. Increase investment in bicycles, as they generate the highest revenue and appear to be the company's core product line.
2. Expand marketing campaigns targeting adults, who represent the largest revenue-generating customer segment.
3. Improve sales of accessories and clothing through bundled promotions with bicycle purchases.
4. Investigate the slight decline in revenue during 2016 to determine whether it resulted from market conditions, operational challenges, or incomplete data.
5. Study high-performing countries to identify successful sales strategies that can be implemented in lower-performing regions.
6. Monitor high-cost products to ensure their selling prices maintain healthy profit margins.
7. Continue using interactive dashboards to support timely business decisions by enabling users to filter data by country, product category, and age group.

## **6. Conclusion**

The Power BI dashboard provides a comprehensive overview of bike sales performance between 2011 and 2016. The analysis indicates strong overall profitability, with bicycles serving as the primary revenue driver and adults forming the largest customer segment. Geographic differences in profitability reveal opportunities for market expansion and targeted improvement. Overall, the dashboard demonstrates how data visualization can support informed decision-making by highlighting key trends, performance indicators, and areas requiring strategic attention.